Session 1: Hardware Basics
Computers and Generations of Computers, Motherboard and its types, Motherboard Components and Architecture, Motherboard Form Factors, Processor and its types

Session 2: Devices
Cabinets and its types, Booting – Warm and Cold, Monitor, Printer, Scanner and its types, SMPS

Session 3: Storage
ROM and RAM, Static and Dynamic RAM, Primary and Secondary Storage, Overview of CMOS and BIOS Setting

Session 4: Printer and Scanner
Installing and Configuring Printer and Scanner

Session 5: OS – Windows 10 or Windows 11
Operating System Functions, Command Prompt Procedures, Major System Files and their Purpose, Managing Files and Directories, Managing Disks

Session 6: OS
Installing and Upgrading Windows, Boot Sequences and Methods, Loading and Configuring Device Drivers, Working with Applications

Session 7: Windows Server OS
Installing and Configuring Windows Server 2019/2022, Installing and Configuring ADDS or Domain Controller, Create Domain Users, Groups and Group Policies

Session 8: Kali Linux or Parrot OS
Installing and Configuring Ubuntu / Parrot OS / Kali Linux, Basic Linux Commands

Session 9: Troubleshooting
Common PC Troubleshooting Commands, Disassembling and Assembling System

Session 10: Networking Fundamentals
Networks and its Types, Modes of Communication & NIC, Networking Cables, Types of LAN Cables & Color Coding

Session 11: Networking Fundamentals
OSI and TCP/IP Model, Networking Topology, Networking Devices, Networking Ports & Protocols

Session 12: Addressing
Addressing (IPv4 & IPv6), LAN Troubleshooting

Session 13: Subnetting
Subnetting (FLSM & VLSM)

Session 14: Networking Fundamentals
Networking Devices, Routers, Layer 2 Switches, Layer 3 Switches, Next-Generation Firewalls, IPS, Access Points, Controllers – Cisco DNA Center, WLC, Endpoints, Servers, Types of Networks, Types of Topology, Two-Tier Architecture, Three-Tier Architecture, Spine-Leaf Architecture, WAN, SOHO, Virtualization Fundamentals, Virtual Machines

Session 15: Networking Fundamentals
OSI Model, TCP/IP Model, Networking Topology, Networking Devices, Networking Ports, Networking Protocols

Session 16: Addressing
IPv4 Addressing, IPv6 Addressing, LAN Troubleshooting

Session 17: Subnetting
FLSM, VLSM

Session 18: Introduction to Routers
Types of Routers, Cisco Hierarchical Design, Internal Components of Routers, External Components of Routers, Router Boot Sequence, Router Modes, Basic Routing Configuration, Basic Commands, Verification Commands

Session 19: Router Security
Password Settings, Password Types, Password Recovery, Banner Mode, SSH Configuration

Session 20: Routing
Static Routing Configuration, Default Routing Configuration

Session 21: Dynamic Routing
Dynamic Routing Protocols – RIP, EIGRP

Session 22: OSPF
OSPF Tables, OSPF Packets, OSPF Router States, WCM Concept, Single Area OSPF Configuration, Multi-Area OSPF Configuration, VLSM with OSPF, BGP Overview

Session 23: Cisco IOS Backup and Recovery
Configuration Backup using TFTP

Session 24: Switching
Types of Switching, Hierarchical Design, Password Configuration, MAC Address Learning, Frame Switching, Flooding, MAC Address Table

Session 25: VLAN and VTP
Types of VLAN, VLAN Configuration, Trunking Configuration, VTP Modes, VTP Configuration

Session 26: Advanced Switching
Port Security Configuration, STP Configuration, RSTP Configuration, EtherChannel Configuration using LACP, CDP, LLDP

Session 27: Inter-VLAN Routing
Layer 3 Switch Configuration, Inter-VLAN Routing using Router

Session 28: Security Fundamentals
Types of ACL, Deny Host to Network, Deny Network to Host, Deny Network to Network, HTTP Configuration, FTP Configuration, ICMP Configuration

Session 29: IPS and NAT
Types of NAT, Static NAT Configuration, Dynamic NAT Configuration, PAT Configuration

Session 30: IP Services
NTP Configuration, DHCP Configuration, DHCP Relay Configuration, DNS Configuration, Syslog Configuration, SNMP Configuration

Session 31: IP Services
TFTP, FTP, HTTP, HTTPS, HSRP Configuration

Session 32: Security Fundamentals
Key Security Concepts, Device Access Control using Local Passwords, VPN GRE Configuration

Session 33: Security Fundamentals
DHCP Snooping, DHCP Relay, Dynamic ARP Inspection, AAA Configuration, RADIUS, TACACS+

Session 34: Security Fundamentals
802.11 Standards, Authentication Types – WPA, WPA2, WPA3, WLAN Configuration using WPA2-PSK (GUI)

Session 35: IPv6
Types of IPv6 Address, EUI-64 Format, Static Routing Configuration, RIPng Configuration

Session 36: IPv6
EIGRPv6 Configuration, OSPFv3 Configuration, Extended ACL Configuration, DHCP Configuration

Session 37: Automation and Programmability
Traditional Networks, Network Automation, SDN, REST-Based APIs Characteristics

Session 38: Automation and Programmability
Configuration Management – Puppet, Chef, Ansible, JSON Data Interpretation

Session 39:
Real-Time Router and Switch Configuration

Session 40: Test
Final Practical Examination